

Root Finding — Summary

The Problem

Find x^* such that $f(x^*) = 0$. **Existence:** if f is continuous and $f(a) \cdot f(b) < 0$, at least one root lies in $[a, b]$ (Intermediate Value Theorem). **Uniqueness:** guaranteed if f is strictly monotonic on $[a, b]$. **Conditioning:** a simple root ($f'(x^*) \neq 0$) is well-conditioned; a multiple root ($f'(x^*) = 0$) is ill-conditioned.

Methods at a Glance

Method	Order	Typ. iters	f' needed?	Robustness	Best for
Bisection	1	~ 20	no	excellent	guaranteed convergence
Newton–Raphson	2	3–5	yes	fair	fast convergence near root
Secant	1.618	4–6	no	good	no derivative available
Fixed-point	1	many	no	poor	special structured problems

Bisection

Repeatedly halve a bracket $[a, b]$ with $f(a) \cdot f(b) < 0$; keep the half containing the sign change.

Error bound after n iterations: $|x_n - x^*| \leq (b-a)/2^n$. $\Rightarrow$ iterations for tolerance ϵ : $n \geq \log_2((b-a)/\epsilon)$.

Pros: always converges (given a valid bracket); no derivatives; trivial implementation. **Cons:** linear convergence (slow); needs an initial bracket.

Newton–Raphson

Linearise f at the current guess via the tangent line:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Quadratic convergence: $|e_{n+1}| \approx C|e_n|^2 \Rightarrow$ correct digits roughly *double* each iteration.

For systems ($\mathbf{F}(\mathbf{x}) = \mathbf{0}$): $\mathbf{x}_{n+1} = \mathbf{x}_n - J(\mathbf{x}_n)^{-1}\mathbf{F}(\mathbf{x}_n)$. In practice, solve $J\boldsymbol{\delta} = -\mathbf{F}$ for $\boldsymbol{\delta}$ using standard method of linear algebra (e.g., LU decomposition, `numpy.linalg.solve`) and then set $\mathbf{x}_{n+1} = \mathbf{x}_n + \boldsymbol{\delta}$.

Failure modes: $f'(x_n) \approx 0$ (huge step); poor initial guess (divergence/oscillation); multiple roots (degrades to linear).

If f' is unavailable: use AD, or approximate with a central difference $f'(x) \approx [f(x+h) - f(x-h)]/(2h)$, $h \sim \sqrt{\epsilon_{\text{mach}}} \sim 10^{-8}$.

Secant Method

Replace $f'(x_n)$ in Newton's formula with a finite-difference approximation using the two most recent iterates:

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

Order $\phi = (1 + \sqrt{5})/2 \approx 1.618$ (golden ratio): superlinear, between bisection and Newton.

Pros: no derivative needed; nearly as fast as Newton. **Cons:** needs two starting points; can fail if $f(x_n) \approx f(x_{n-1})$.

Fixed-Point Iteration

Rewrite $f(x) = 0$ as $x = g(x)$ and iterate $x_{n+1} = g(x_n)$.

Convergence condition: $|g'(x^*)| < 1$. Rate is linear: $|e_{n+1}| \approx |g'(x^*)| |e_n|$.

Common choice: $g(x) = x - \lambda f(x)$; pick λ so that $|g'(x^*)| = |1 - \lambda f'(x^*)| < 1$.

Note: Newton's method is the fixed-point iteration $g(x) = x - f(x)/f'(x)$, which achieves $g'(x^*) = 0$ at a simple root $\Rightarrow$ quadratic convergence.

Error Analysis and Stopping Criteria

Two distinct measures:

- **Absolute error:** $e_n = |x_n - x^*|$ (unknown in practice).
- **Residual:** $r_n = |f(x_n)|$ (computable, but small residual $\nRightarrow$ small error if $f'(x^*) \approx 0$).

Recommended stopping criterion (combined):

$$|f(x_n)| < \epsilon_1 \quad \text{AND} \quad |x_n - x_{n-1}| < \epsilon_2.$$

Convergence order: a sequence converges with order p if $\lim_{n \rightarrow \infty} |e_{n+1}|/|e_n|^p = C < \infty$.

Practical Strategies

Hybrid bisection + Newton (recommended)

1. Run bisection for 5–10 iterations to get a safe initial guess near the root.
2. Switch to Newton (or secant) for fast final convergence.
3. Combines robustness of bisection with speed of Newton.

Choosing a method

Situation	Recommendation
Need guaranteed convergence	Bisection
Have f' and good initial guess	Newton–Raphson
No derivative available	Secant method
Best speed + robustness	Hybrid (bisection $\rightarrow$ Newton)
System of equations $\mathbf{F}(\mathbf{x}) = \mathbf{0}$	Newton with Jacobian (or quasi-Newton)

Checklist

- Plot $f(x)$ first to locate approximate roots and detect multiple roots / near-singularities.
- Verify bracket sign change (bisection) or check $|g'(x^*)| < 1$ (fixed-point).
- Use *both* residual and iterate-change as stopping criteria.
- For Newton on systems: solve $J\delta = -\mathbf{F}$ (never invert J explicitly).
- Test on simple functions with known roots before applying to real problems.